INFERENCE (the student alone drives)

Instruction

"Walk past the sofa, then stop at the bedroom door."

same text as NaVILA's prompt

Observations: 8 frames, sampled as NaVILA does



7 past frames, evenly spaced from the start



current frame

Laya question layout

ModernBERT tokenizer, ≤ 448 tokens

question · 10 options · state {instruction}

SigLIP2-B/16 vision encoder

224 px, shared by all frames

92.9M, pretrained

current: 14×14 = 196

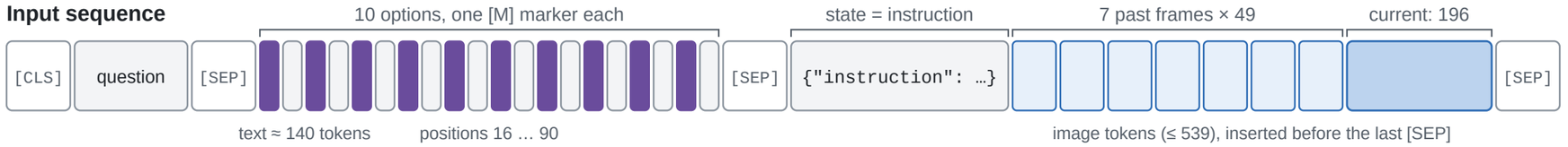
past: pool → 7×7 = 49 each

Projection MLP 768 → 1024

+ frame-slot emb. (slot 7 = current)

+ 2-D position emb. · new, 2.1M

Input sequence



text ≈ 140 tokens · positions 16 … 90

image tokens (≤ 539), inserted before the last [SEP]

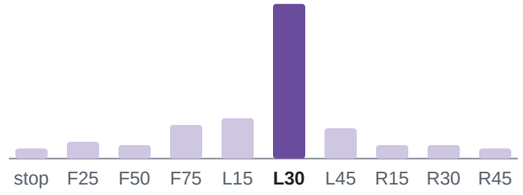
ModernBERT-large encoder 28 layers · d = 1024 · RoPE · 394.8M · pretrained (Laya checkpoint or raw ModernBERT)



Laya decision head 2 Transformer layers, full attention, 16 heads · 25.2M · read out only at the 10 [M] positions

scorer MLP: one logit per [M] · 1.05M

softmax over NaVILA's 10 answers



argmax → NaVILA sentence

"The next action is turn left 30 degree."

NaVILA executor (unchanged)

→ 2 × turn-left 15°

Habitat VLN-CE / Go2 locomotion

516.3M parameters = encoder 394.8M + head and scorer 26.5M + vision 92.9M + new 2.1M

NaVILA 8B: 15.5× more. One forward pass per decision, no token decoding · 54.8 ms median on an RTX 5060 Ti (R2R-CE)

DISTILLATION: NaVILA 8B → student

teacher distillation + auxiliary tasks · not used when driving

Teacher: NaVILA 8B (frozen)

Scores all 10 answer sentences on one KV-cached prompt: same 8 frames and instruction → p_teacher over the 10 answers. Reads the instruction, so the 3 instructions of one R2R path get different distributions.

1 · Navigation decision (80%)

target A: one-hot NaVILA label

target B: (1 − α)·one-hot + α·p_teacher, α = 0.5

over the 10 [M] options

left–right mirror with p = 0.5 (frames, text, label)

2 · Instruction–route matching (20%)

same model, same layout, whole route

Q: "Does the instruction describe this route?"

options: yes / no (8 unused markers masked)

negatives:

· landmark swap: sofa → bed (only the frames tell)

· instruction of another path

why: R2R labels are identical for the 3 instructions of a path, so labels alone never reward reading the instruction

Loss: Laya RLCD

soft cross-entropy + policy gradient on noisy logits, σ 0.4 → 0.1

same loss for both tasks (masked options)

Legend

- pretrained vision (SigLIP2)
- pretrained language encoder
- Laya head, scorer, [M] markers
- new parameters (2.1M)
- text tokens
- distillation (training only)